

Porting AI/ML Models to Intelligence Processing Units (IPUs)

Abhinand S. Nasari[†], Lujun Zhai[†], Zhenhua He, Hieu T. Le, Jian Tao, Suxia Cui,
Dhruva K. Chakravorty, Lisa M. Perez and Honggao Liu

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PEARC23

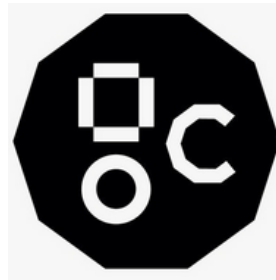


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Introduction and Motivation

- Purpose-built accelerators for AI/ML applications offer opportunities to speed up the training and inference.
- NSF ACES cyberinfrastructure (CI) testbed



BOW IPU PROCESSOR

Deep Trench Capacitor

Efficient power delivery
Enables increase in operational performance

Wafer-On-Wafer

Advanced silicon 3D
stacking technology
Closely coupled power
delivery die
Higher operating frequency
and enhanced overall
performance

IPU-Tiles™

1472 independent IPU-Tiles™ each with an
IPU-Core™ and In-Processor-Memory™

IPU-Core™

1472 independent IPU-Core™

8832 independent program threads
executing in parallel

In-Processor-Memory™

900MB In-Processor-Memory™ per IPU

65.4TB/s memory bandwidth per IPU

Solder Bumps

IPU-Links™

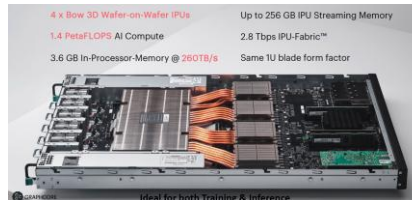
10x IPU-Links,
320GB/s chip to chip bandwidth

IPU-Exchange™

11 TB/s all to all IPU-Exchange™
Non-blocking, any communication pattern

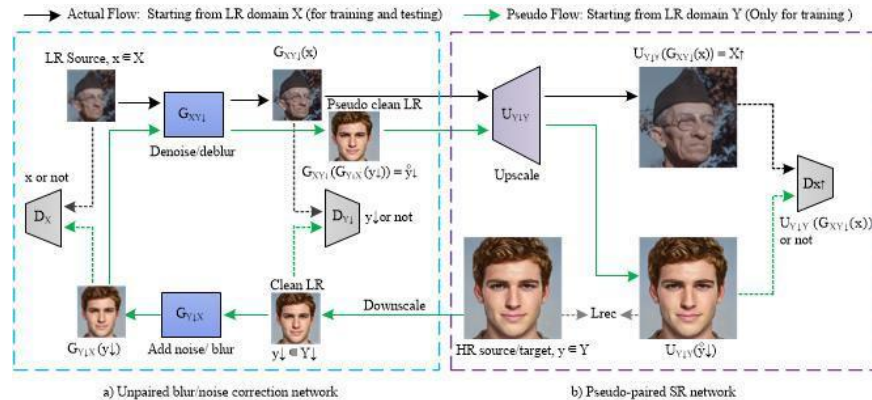
PCIe

PCI Gen4 x16
64 GB/s bidirectional bandwidth to host



TransCycleGAN Network

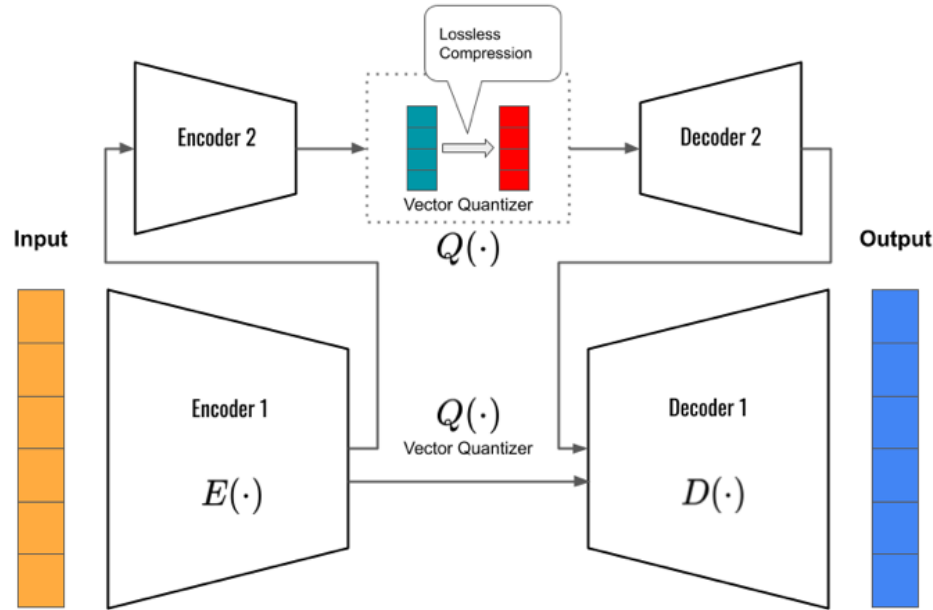
The TransCycleGAN network consists of an unpaired blur/noise correction network integrated with a degradation-removal Transformer in the Low-resolution domain and a pseudo-paired SR network for reconstructing High-resolution from a low resolution image.



Quality Comparison of the Images Produced by Various Super-Resolution Models

Method	PSNR	SSIM
SRGAN [24]	26.02	0.7397
EDSR [25]	28.80	0.7876
RCAN [26]	28.87	0.7889
HAT-L [27]	29.47	0.8015
TranCycleGAN	29.78	0.8103

Hierarchical Autoencoder

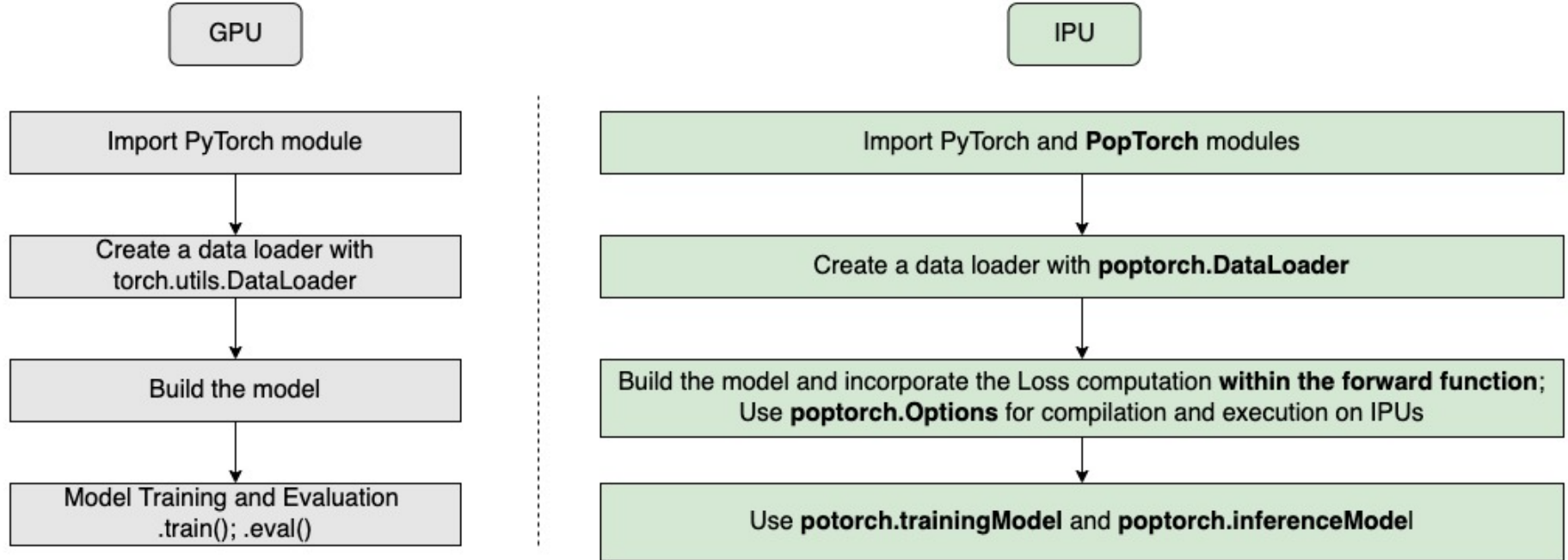


The architecture of Hierarchical Autoencoder. The input of the neural network is the original scientific data, and the output is the reconstructed data. In the ideal case, the input and output data are identical. The compression ratio achieved by the neural network is calculated by dividing the size of the input data (represented with orange blocks) by the size of the compressed latent space (represented by red blocks).

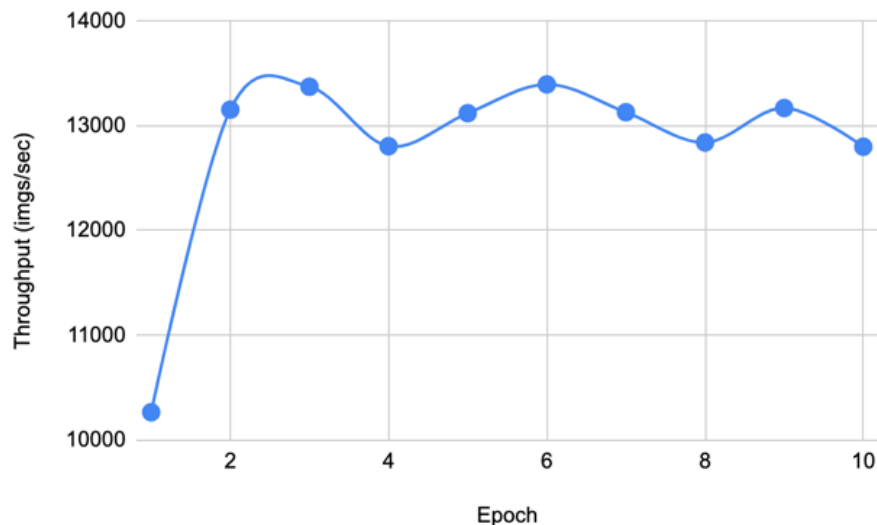
Test Environment

- ACES testbed at the West Campus Data Center at Texas A&M University
- 16 Colossus GC200 IPUUs and 16 Bow IPUUs
- Graphcore GitHub repository: various deep learning models for benchmarking purposes
- Poplar Software Development Kit (SDK) 3.2.0
- PyTorch (PopTorch 3.2.0) framework

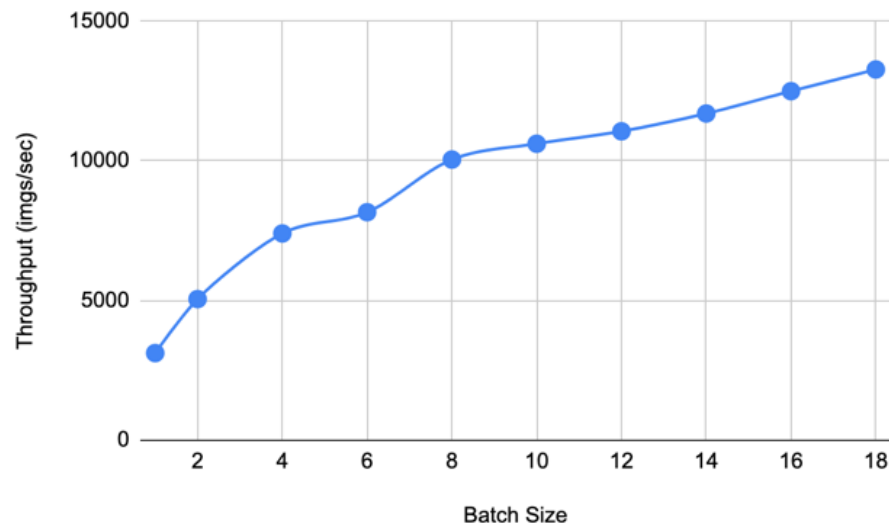
Converting PyTorch models to PopTorch for IPUs



Performance of IPUs on ACES for PyTorch ResNet50

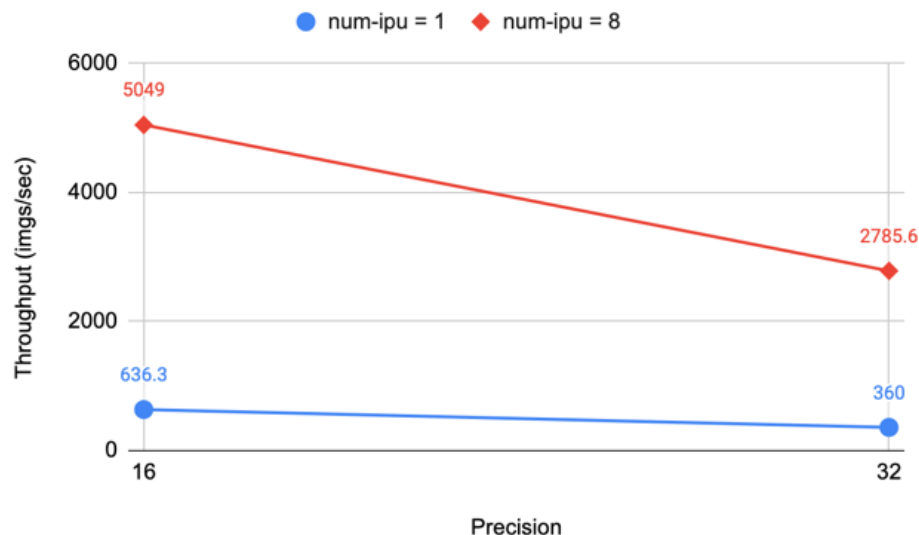


The image throughput with the number of epochs over 8 IPUs. The first epoch has lower performance due to graph compilation. The performance stabilizes after the second epoch.

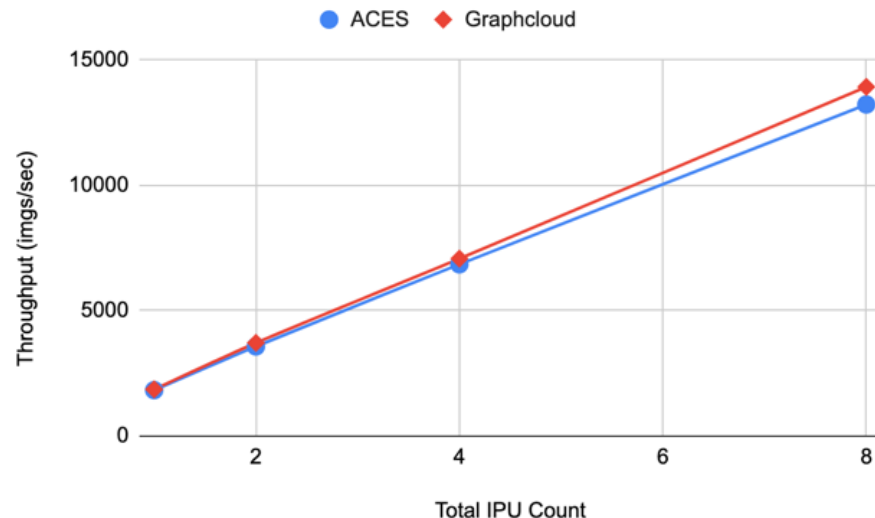


The image throughput of PyTorch ResNet 50 model increases with batch size. Smaller batch sizes (less than 4) had a greater increase in image throughput with batch size compared to larger batch sizes (greater than 8).

Performance of IPUs on ACES for PyTorch ResNet50



Impact of precision on the image throughput of PyTorch ResNet 50 model for 1 and 8 IPUs. Batch size was set at 2., and a similar performance drop (~44%) was observed in switching from half to full precision, for both 1 and 8 IPUs.



The image throughput increases almost linearly with the number of IPUs on the NSF ACES cluster (blue line) and the Graphcloud service (red line). To ensure maximum throughput, these calculations were performed using a batch size of 16 with half precision.

Training Performance of IPUUs on the TransCycleGAN

Accelerator	Count	Images/sec
IPU	1	438
IPU	16	6552

Performance of IPUs on the Hierarchical Autoencoder

Accelerator	Stage	Count	Images/sec
IPU	Training	1	1.69
IPU	Training	16	16.58
IPU	Inference	1	2.0

Conclusions

- PyTorch ResNet50 model has an almost linear scaling behavior on the IPU's on the ACES system, ensuring that researchers will experience the same performance characteristics as on the Graphcloud cloud service.
- The TransCycleGAN and Hierarchical Autoencoder models in PyTorch framework have been successfully ported to IPU's.
- Our study evaluated the performance of IPU's on the TransCycleGAN and Hierarchical Autoencoder models and found that increasing the number of IPU's significantly improved the model's throughput.

Acknowledgements

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